

THE UNIVERSITY OF HONG KONG
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE

Final Exam

COMP3314 Machine Learning

17 December 2021, 9:30 am - 12:00 noon

UNIVERSITY NO. : _____

(DO NOT WRITE YOUR NAME)

Candidates are permitted to bring to the examination hardcopies of the lecture slides of the course provided by the course instructor, on which handwritten notes are allowed. There are 6 questions to be answered. Write your answers in the spaces provided in the exam paper. Write your University No. on every page, do not write your name.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and Model of the calculator: _____

Question	Max. Mark	Your Mark (examiner use only)
1	9	
2	20	
3	15	
4	18	
5	20	
6	18	
Total	100	

1. Multiple Choices

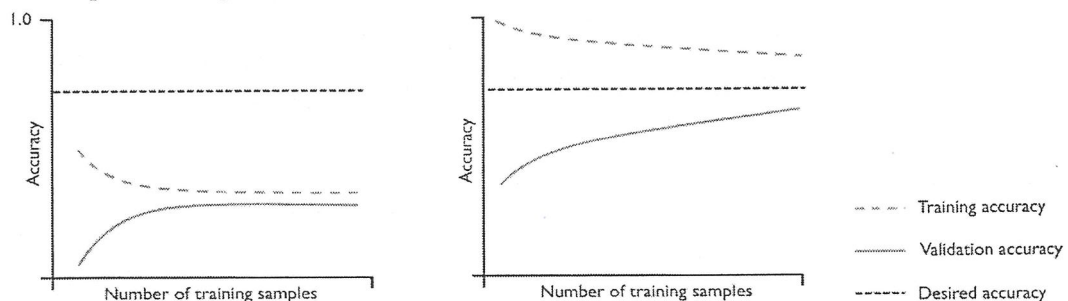
1.1 (3 points): Select the correct answer. What are the axes of an ROC curve?

- A. Vertical axis: % of true negatives; Horizontal axis: % of false negatives
- B. Vertical axis: % of true positives; Horizontal axis: % of false positives
- C. Vertical axis: % of false negatives; Horizontal axis: % of false positives
- D. Vertical axis: % of false positives; Horizontal axis: % of true negatives
- E. None of the above

1.2 (3 points): In a medical application domain, suppose we build a classifier for patient screening (True means patient has cancer). Which of the following situations would you like your classifier to have? Select ALL correct answers. (FP: false positive, FN: false negative, TP: true positive, TN: true negative)

- A. $TP \gg FN$
- B. $FN \gg FP$
- C. $FN = FP \times TP$
- D. $FP \gg FN$
- E. $FN \times TP \gg FP \times TN$

1.3 (3 points): Consider the following two learning curve plots. Label them to indicate which plot corresponds to the listed characteristics:



- | | |
|------------------------|----------------------|
| A. Left: High variance | Right: High bias |
| B. Left: Low variance | Right: High bias |
| C. Left: Low bias | Right: High variance |
| D. Left: High bias | Right: High variance |

2. Short Answer

2.1 (5 points): In feature scaling, what is the difference between standardization and normalization?

2.2 (5 points): Explain confusion matrix, precision, recall and F1 score.

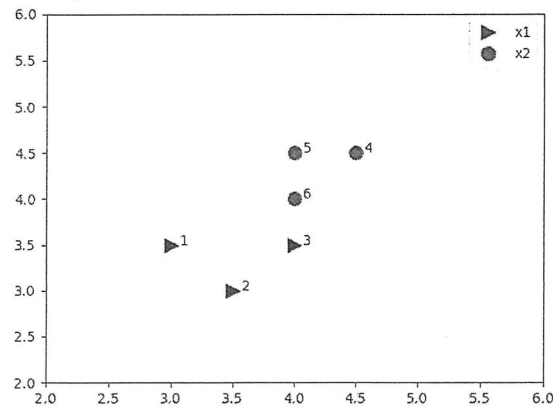
2.3 (5 points): Class imbalance can be a problem in machine learning. Explain why and describe two strategies to tackle this problem.

2.4 (5 points): What is k-fold cross-validation? Should k increase or decrease when the dataset is small? Why?

3. Logistic Regression

We attempt to solve the binary classification task depicted in the figure below with a simple linear logistic regression model (x_1, x_2 are respectively horizontal and vertical coordinates in the figure):

$$P(y = 1 | \vec{x}, \vec{w}) = \varphi(w_0 + w_1 x_1 + w_2 x_2) = \frac{1}{1 + \exp(-w_0 - w_1 x_1 - w_2 x_2)}$$



3.1 (9 points): Notice that the training data can be separated with zero training error with a linear separator. Consider training *regularized* linear logistic regression models where we try to maximize

$$\sum_{i=1}^n \log(P(y_i | x_i, w_0, w_1, w_2)) - C w_j^2$$

for very large C . The regularization penalties used in penalized conditional loglikelihood estimation are $-C w_j^2$, where $j = \{0, 1, 2\}$. In other words, only one of the parameters is regularized in each case. Given the training data in the figure above, how does the training error change with regularization of each parameter w_j ? State whether the training error increases or stays the same (zero) for each w_j for very large C . Provide a brief justification for each of your answers. Note: no calculation is needed.

- By regularizing w_2 .
- By regularizing w_1 .
- By regularizing w_0 .

3.2 (6 points): Given the loss function defined as $J^i(\mathbf{w})$, suppose we use Stochastic Gradient Descent, please derive $\frac{\partial J^i(\mathbf{w})}{\partial \mathbf{w}} = (\varphi - y^i)\mathbf{x}^i$.

$$J^i(\mathbf{w}) = -y^i \log(\varphi(z^i)) - (1 - y^i) \log(1 - \varphi(z^i))$$

where $z^i = \mathbf{w}^T \mathbf{x}^i + b$, $\varphi(z^i)$ is the sigmoid function and y^i is the label for the i -th sample $\mathbf{x}^i$.

4. Linear SVM

Consider a classification problem: there is a one-feature sample and class label y can be 1 or -1. There are 2 data points, which are

$$\begin{aligned}(x_1, y_1) &= (1, 1) \\ (x_2, y_2) &= (-1, -1)\end{aligned}$$

4.1 (6 points): For this problem write down the QP problem for an SVM with slack variables and Hinge loss. Denote the weight for the slack variables ξ_i is C . Let the equation of the decision boundary be

$$wx + b = 0$$

We denote the QP form as:

$$\begin{aligned}\min_{\mathbf{z}} \quad & \frac{1}{2} \mathbf{z}^T \mathbf{P} \mathbf{z} + \mathbf{q}^T \mathbf{z} \\ \text{s. t.} \quad & \mathbf{G} \mathbf{z} \leq \mathbf{h}\end{aligned}$$

Let $\mathbf{z}$ be optimizable variables. QP problem has a quadratic optimization term and linear constraints. Please show that SVM with slack variables and Hinge loss can be considered as a QP form.

4.2 (6 points): Let's assume we add some more data points and have solved the quadratic programming problem. There is a negative support vector at $x = 0$ and a positive support vector at $x = 2$. What is the width of the margin (twice the distance from decision boundary to the closest training samples)? What is the weight w and bias b ?

4.3 (6 points): Let two positive samples at $x = 0.5$ and $x = 3$. What are the slack variables for them?

5. Perceptron

5.1 (4 points): Draw a fully connected network that has 3 neurons in the input layer, 5 neurons in a hidden layer and 2 neurons in the output layer.

5.2 (6 points): Assuming that the activation function is linear, show that this network can be simplified into one without a hidden layer. Note: the output layer does not need an activation function.

5.3 (10 points): With the network defined in 5.1, assume the input layer is X , hidden layer is Y and output layer is Z . Activation function is $ReLU$ function. The overall formulas are defined as:

$$\begin{aligned} Y &= W_1^T X + b_1 \\ Y_{act} &= ReLU(Y) \\ Z &= W_2^T Y_{act} + b_2 \\ l &= Loss(Z) \end{aligned}$$

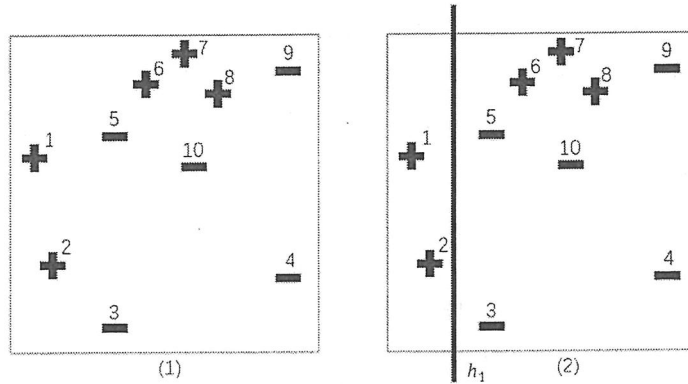
Where X is $[1, 2, 3]$, b_1 is $[-0.5320, -0.5185, -0.3604, -0.5402, -0.4878]$ and $\frac{\partial l}{\partial Y_{act}}$ is $[0.2345, -0.0214, -0.0060, 0.2950, 0.2095]$. Calculate $\frac{\partial l}{\partial X}$ and $\frac{\partial l}{\partial W_1}$.

$$W_1^T$$

0.2085	0.3803	-0.5659
0.4626	0.4808	-0.3905
-0.2183	0.1416	0.0663
0.1253	0.0960	0.5014
0.5484	0.3817	-0.4773

6. Boosting

Consider training a boosting classifier using decision stumps (decision stump refers to one level decision tree) as weak learners on the dataset shown in the following figure (1). Each sample has two features represented as the x and y coordinates in the following 2D plots.



- (a) **(5 points)** Assume at the end of the first iteration, h_1 is the first weak learner obtained. The left side of h_1 is classified as positive and the right side is classified as negative. Which samples will have their weights increased at the end of the first iteration? Circle them.
- (b) **(8 points)** How many iterations will it take to achieve zero training error? Explain.
- (c) **(5 points)** Why do we want to use “weak” learners when boosting?

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